Object Oriented Programming Using C++

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Template

- If we want to write generic code in C++, then we should use template.
- Objective
    1. Not to reduce code size
    2. Not to reduce execution time
    3. To reduce developers effort.
- It is possible by passing data type as a argument.
- Types of template:
    1. Function template
    2. Class Template

Function Template

```
//template<typename T>     //T => Type Parameter.
template<class T>    //T => Type Parameter.
void swap_object( T &obj1, T &obj2 ){
    T temp = obj1;
    obj1 = obj2;
    obj2 = temp;
}
```

```
int main( void ){
    int x = 10;
    int y = 20;
    ::swap_object<int>(x, y);    //OK //int => Type argument
    ::swap_object(x, y);     //OK
    cout<<"X    :    "<<x<<endl;
    cout<<"Y    :    "<<y<<endl;
    return 0;
}
```

Template

- An ability of compiler to detect and pass type of argument implicitly to the function is called type inference.
- Template is mainly designed for data structure and algorithm.
- By passing data type as a argument, we can define generic code in C++. Hence parametrized type is called template.
- We can not divide template code into multiple files.